Risk factors of recurrent thyroid nodules after radiofrequency ablation

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Abstract

Objective: To investigate the risk factors of thyroid nodule recurrence after radiofrequency ablation (RFA). METHODS The medical record information of 120 patients with thyroid nodules admitted to our hospital from June 2019 to April 2022 was retrospectively analysed. All participants received RFA treatment. According to the results of the postoperative thyroid ultrasound examination (USG), the patients were divided into the recurrence group (R, N=16) and the non-recurrence group (NR, N=104). Binary logistic regression analysis was performed to identify the independent risk factors of thyroid nodule recurrence after RFA. The receiver operating characteristic (ROC) curve was used to analyse the value of the forecast of each independent factor and combined model for thyroid nodule recurrence after RFA.

Results: During the follow-up period, 16 patients recurred, and the recurrence rate was 13.33%. Univariate regression analysis showed that whether the nodules are solitary (WNS), nodule diameter (ND), the degree of risk of nodular location (DRN), recurrent laryngeal nerve (RLN) injury were associated with thyroid nodule recurrence after RFA (P<0.05). Binary logistic regression analysis showed that WNS, ND, DRN and RLN injury were independent risk factors for the recurrence of thyroid nodules after RFA (P<0.05). ROC analysis of independent factors and combined model showed that solitary nodules, nodule diameter and nodule location risk degree had diagnostic value, while RLN injury had no predictive value. The combined model is more predictive than the independent factors. Conclusions: The risk factors of recurrent thyroid nodules after radiofrequency ablation are related to WNS, ND, DRN and so on, which should be paid attention to and preventive measures should be taken.

Keywords: Thyroid nodules; radiofrequency ablation; recurrence; risk factors.

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Introduction

A thyroid nodule is a mass inside the thyroid gland that moves up and down with the thyroid gland during swallowing. The disease is very common in clinical treatment, and the incidence of women is significantly higher than that of men. It is shown that 90% of the detected nodules are benign nodules [1, 2]. Most benign nodules can be followed up without treatment. When the symptoms of neck compression, combined with hyperthyroidism and medical treatment ineffective or malignant tendency, surgical resection is the traditional treatment [3]. However, the disadvantages of total thyroidectomy are very obvious. This type of surgery can cause great trauma to the patient's body in the process of treatment. The postoperative scar is easy to leave, not only slowing down the postoperative recovery of patients but also affecting the appearance. Normal thyroid glands may be damaged during the operation, resulting in adverse effects of treatment [4]. In addition, the residual gland recurrence rate is still high after total or partial thyroidectomy or nodule excision. And it is prone to different degrees of hypothyroidism complications [5].

Ultrasound-guided RFA is gradually applied to patients with recurrent thyroid nodules and thyroid cancer after surgery. Under the guidance of ultrasound, high-frequency alternating current is generated through radiofrequency electrodes, which cause polar molecules and electrons

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in the tissue to oscillate at high speed, thereby generating heat 6. When the temperature approaches 50 °C, the protein solidifies, leading to necrosis of the tumor cells and thus elimination of the tumor [7]. This surgical method has the advantages of small trauma, quick recovery, high safety and good efficacy, so it is often used in the clinical treatment of tumors and nodules [6, 8].

The recurrence rate of thyroid nodules after surgery is high. Studies in China reported that the recurrence rate was 18-30% 9, while other countries reported that the recurrence rate was 10-45% 10, and most of them recurred within 5 years after surgery. The high recurrence rate harms the prognosis of patients. To further understand the risk factors affecting their recurrence, we analysed 120 patients with thyroid nodules who underwent radiofrequency ablation treatment at our hospital between June 2019 and April 2022. They were followed up after RFA treatment and compared the clinical data of patients with and without recurrence after surgery. To provide a clinical reference for ultrasound-guided RFA in the treatment of thyroid nodules.

Methods

Patients

A total of 120 patients with thyroid nodules admitted to our hospital from June 2019 to April 2022 were included in this study. All participants were treated with RFA after admission.

Inclusion criteria: (1) All patients showed obvious clinical symptoms and thyroid nodules were found during physical examination or palpation; (2) Preoperative histological biopsy showed benign nodules; (3) Preoperative thyroid function was normal; (4) The coagulation function was normal and the liver and kidney functions were normal. (5) The treatment cooperation of the patients was good, and the patients and their relatives were aware and agree and signed informed consent.

Exclusion criteria: (1) Patients with vital organ dysfunction, including cardiopulmonary function, liver and kidney function impairment; (2) Previous history of neck surgery or radiotherapy; (3) Patients with poor compliance without regular follow-up in our hospital.

Follow-up

The thyroid function and ultrasound examination (USG) were reviewed every 3 months after the operation. If the condition is relatively stable for more than 1 year, it can be reviewed every 6 months. The starting point of follow-up was the operation time of the patient, and the end point of follow-up was the recurrence of the patient or the last follow-up time (July 2022). Recurrence was defined as a USG-suggestive nodule more than 3 months after initial treatment. Data were collected including gender, age, and grouping according to recurrence.

Statistical analysis

Statistical analysis was conducted by SPSS version 23 software. The continuous variables were displayed as mean ± standard deviation. Count variables were represented by n (%). A Student's t-test and x^2 test were performed to evaluate the correlation between demographic and preoperative and postoperative characteristics. Binary logistic regression was used to analyse the risk factors of recurrent thyroid nodules after RFA treatment. Univariate analysis was used to obtain meaningful variables, and then the meaningful variables were included in binary regression model analysis to screen out the independent risk factors of recurrent thyroid nodules after RFA. The ability of each independent risk factor and combined model to predict thyroid recurrence was analysed by receiver Operating characteristic curve (ROC) analysis tool of SPSS version 23 software. The ROC curve is a curve reflecting the relationship between sensitivity and specificity. Area Under Curve (AUC), which is used to represent the prediction accuracy. The AUC value is proportional to the accuracy. It is generally believed that 0.5 <AUC≤0.7 suggests poor prediction ability, 0.7 <AUC≤0.9 implies better predictive ability, and AUC> 0.9 indicates high predictive value. P <0.05 was considered a statistically significant difference. /span>

Results

The male (n=31, 25.83%) to female (n=89, 74.17%) ratio is about 1:3, with an average age of 52.55±7.03 years, ranging from 35 to 72 years. There were 48 patients with multiple nodules and 72 patients with solitary nodules, with a total of 148 nodules. There were 95 solid nodules and 53 cystic nodules, with an average diameter of 1.94±0.43cm (1.06-2.99 cm). The number of cystic and solid nodules was 39 and 81, respectively. Up to the last follow-up in July 2022, the follow-up time was 3-35 months. During the follow-up period, 16 cases (13.33%) had a recurrence. The demographic characteristics of the recurrence group (R) and the non-recurrence group (NR)

were compared. As shown in Table 1, statistical differences were found in the proportion of whether the nodules are solitary (WNS), nodule diameter (ND), nodule components, the degree of risk of nodular location (DRN) and whether recurrent laryngeal nerve (RLN) injury occurred in the two groups (P<0.05). No significant differences were found in gender, age and aspects between the two groups (P>0.05). Table 2 shows the preoperative thyroid function test showed that the patients were normal, and the FT3, FT4 increased slightly at 6 months after the operation, but there was no statistical significance between R and NR (P>0.05).

Table 1: Demographic characteristics

Variable	Total (n=120)	NR(n=104)	R(n=16)	χ^2	P
Gender					
Male	31(25.83%)	29(27.88%)	2(12.50%)	1.004	0.316
Female	89(74.17%)	75(72.12%)	14(87.50%)		
Age(year)					
<60	103(85.83%)	90(86.54%)	13(81.25%)	0.032	0.857
≥60	17(14.17%)	14(13.46%)	3(18.75%)		
Solitary nodules					
No	32(26.67%)	22(21.15%)	10(62.50%)	3.894	0.048
Yes	88(73.33%)	82(78.85%)	6(37.50%)		
Average diameter of nodules (CM)					
<2	74(61.67%)	69(66.35%)	5(31.25%)	7.225	0.007
≥2	46(38.33%)	35(33.65%)	11(68.75%)		
Nodule component					
solid component <20%	41(34.17%)	32(30.77%)	9(56.25%)	4.003	0.045
solid component ≥20%	79(65.83)	72(69.23%)	7(43.75%)		
Risk of nodular location					
Low risk	96(80.00%)	90(86.54%)	6(37.50%)	20.841	<0.001
High risk	24(20.00%)	14(13.46%)	10(62.50%)		
Recurrent laryngeal nerve injury					
No	106(88.33%)	96(92.31%)	10(62.50%)	11.955	0.001
Yes	14(11.67%)	8(7.69%)	6(37.50%)		

Table 2: Preoperative and postoperative thyroid function

Measurements		Total (n=120)	NR(n=104)	R(n=16)	t	P
Preoperative thyroid function	FT3 (pmol/L)	4.87±0.52	4.86±0.54	4.93±0.41	-0.512	0.610
	FT4 (pmol/L)	16.51±1.82	16.17±1.22	15.55±1.39	1.843	0.068
	TSH (mU/L)	1.78±0.57	2.24±0.63	1.96±0.76	1.585	0.116
Thyroid function at 6 months after surgery	FT3 (pmol/L)	5.11±0.71	5.07±0.7	5.33±0.78	0.176	0.176
	FT4 (pmol/L)	16.82±2.21	16.88±2.19	16.36±2.4	0.875	0.383
	TSH (mU/L)	1.85±0.65	1.85±0.67	1.82±0.48	0.166	0.868

Of the 16 recurrent patients, 10 (62.5%) had nodules in high-risk locations, compared with 14 (13.46%) in the NR group. The incidence of RLN injury in the NR group was 8.51%, and all 6 cases were unilateral RLN injury. Among them, 4 cases presented with choking in drinking water without treatment, which improved spontaneously one week after operation, and the other 4 cases presented with hoarseness, which recovered to normal 5 days after nutritional nerve treatment. The incidence of postoperative RLN injury in group R was 37.5%, and all 6 cases were unilateral RLN injuries, manifested as postoperative hoarseness, which was treated with nutritional nerve therapy and returned to normal on the fifth day after operation.

A binary logistic regression model was established, with all patients as samples and recurrence as the dependent variable, with 1= recurrence and 0= no recurrence. The independent variables were gender, age, WNS, ND≥2cm, nodule composition, DRN, and RLN injury, and the assignment table was shown in Table 3. The input method was used for univariate regression analysis, and the results are shown in Table 4. WNS, ND, DRN and RLN injury were risk factors for thyroid nodule recurrence (P<0.05). There is no statistical difference between thyroid nodule reoccurrence and gender, age ≥60 years, and nodule composition (P>0.05).

Table 3: The assignment of variables

Variable	Assignment
Whether recurrence	No=0, Yes=1
Gender	Male=0, Female=1
Age(year)	<60 =0, ≥60 =1
Solitary nodules	No=0, Yes=1
Average diameter of nodules (CM)	<2=0, ≥2=1
Nodule component	solid component <20%=0, solid component ≥20%=1
Risk of nodular location	Low risk=0, High risk=1
Recurrent laryngeal nerve injury	No=0, Yes=1

Table 4: Univariate logistic regression analysis

Variable	β	Standard error	Waldχ^2	P	95%CI
Gender	0.996	0.787	1.601	0.206	2.707(0.579,12.655)
Age	0.394	0.702	0.316	0.574	1.484(0.375,5.873)
Solitary nodules	-1.827	0.569	10.287	0.001	0.161(0.053,0.491)
Average diameter of nodules (CM)	1.467	0.578	6.446	0.011	4.337(1.397,13.462)
Nodule component	-1.062	0.547	3.772	0.052	0.346(0.118,1.01)
Risk of nodular location	2.372	0.591	16.106	<0.001	10.714(3.365,34.117)
Recurrent laryngeal nerve injury	1.974	0.634	9.692	0.002	7.200(2.078,24.951)

According to the results of univariate regression analysis, WNS, ND, DRN and RLN injuries were included in the binary regression model. The analysis results were displayed in Table 5. WNS, ND, DRN and RLN injuries were independent risk factors for thyroid nodule recurrence (P<0.05). Hosmer-Lemeshow goodness of fit test showed that X^2 =5.575, P=0.233, indicating good fit and high reliability of the model.

Table 5: Binary logistic regression analysis

Variable	β	Standard error	Waldχ^2	P	95%CI
Solitary nodules	-2.757	0.935	8.689	0.003	0.063(0.010,0.397)
Average diameter of nodules	1.746	0.821	4.525	0.033	5.732(1.147,28.641)
Risk of nodular location	3.793	1.051	13.022	<0.001	44.391(5.657,348.337)
Recurrent laryngeal nerve injury	3.928	1.224	10.303	0.001	50.806(4.616,559.172)

ROC curve analysis of the above independent risk factors and their combined model showed that the AUC of each factor was 0.707 for WNS, 0.675 for ND, 0.745 for DRN, and 0.924 for combined model (95%CI: 0.847-1.000), with statistical significance (P<0.05). The AUC of RLN injury was 0.649, and there was no statistical significance (P>0.05). ND was greater than 0.5, but less than 0.7 (P=0.024), indicating that ND has a significant diagnostic value for thyroid nodule recurrence after RFA, but its diagnostic value is still low. The combined value of independent risk factors was higher than that of single risk factors. It indicates that the model has good discrimination and high diagnostic value, as shown in Figure 1 and Table 6.

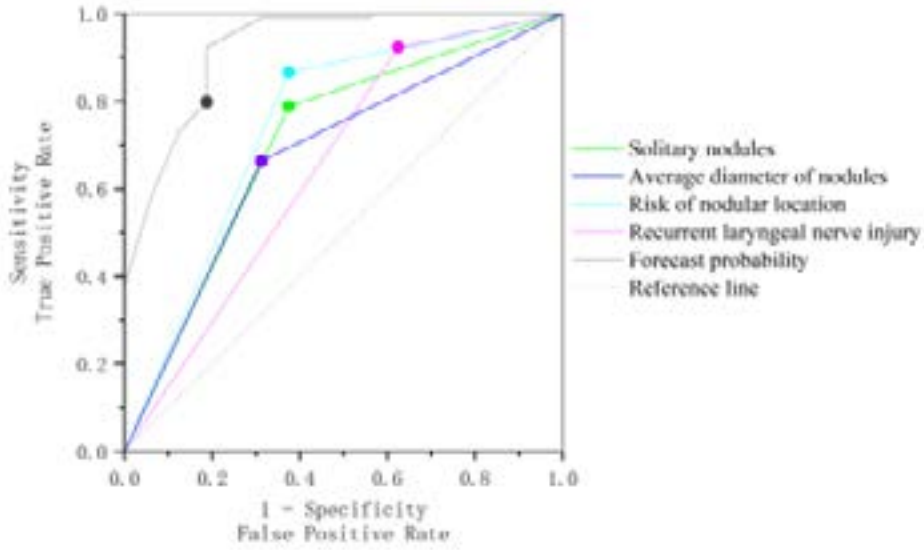


Figure1: Receiver operating curves of independent factors and combined model

Table 6: Receiver operating curve and area under curve (AUC)

Test result variable	AUC	P	Sensitivity	Specificity	95%CI
Solitary nodules	0.707	0.008	0.625	0.788	(0.560,0.854)
Average diameter of nodules	0.675	0.024	0.688	0.663	(0.534,0.817)
Risk of nodular location	0.745	0.002	0.625	0.865	(0.598,0.892)
Recurrent laryngeal nerve injury	0.649	0.056	0.375	0.923	(0.486,0.813)
Forecast probability	0.924	<0.001	0.813	0.923	(0.847,1.000)

Discussion

Although thyroid nodules are mostly benign lesions, nodules will affect the aesthetics of patients, and the large volume of nodules will cause local compression. Therefore, patients with thyroid nodules need to be treated effectively as early as possible to improve their quality of life. At present, RFA therapy is mainly used in the clinical treatment of tumor lesions. It has been reported that the 5-year recurrence rate of thyroid RFA is 20%-23%[11, 12]. Its pathogenesis is diffuse thyroid hyperplasia and enlargement caused by long-term iodine deficiency, and long-term proliferative and degenerative lesions alternate repeatedly, leading to nodules. However, in a small number of cases, nodule regrowth is caused by an underlying malignant tumor [13]. Li et al. [14] reported that RFA had a low effective rate in the treatment in children, with a volume reduction rate (VPR) of 77.5% and a nodule regeneration rate of 22.9% at 12 months after operation. Du et al. [15] applied microwave ablation in patients with thyroid nodules and found that nodules gradually decreased with time through regular follow-up, and the VPR of nodules was 96.9% at 48 months after operation. Although the efficacy of RFA is good, postoperative disease recurrence can affect the efficacy of treatment [6].

Due to the relatively complex relationship between the tissues around the thyroid gland, there are many blood vessels and nerves distributed, which increases the risk of surgery to a certain extent. The use of expanded ablation treatment is prone to cause thermal damage to the blood vessels and nerves of patients, resulting in articulation disorders, nodule rupture and other complications [16,17]. The incidence of major complications in the application of RFA in the thyroid nodule has been reported to be as high as 3.8% [18], and another study of RFA applied to the thyroid nodules in children reported complication rates as high as 4.8% and found higher VPR in nodules with a predominantly cystic component [14]. Therefore, it makes sense explore the risk factors of thyroid nodule recurrence after RFA and take corresponding intervention measures to improve the prognosis of the disease.

Our analysis showed that WNS, ND, DRN and RLN injury were all risk factors for the recurrence of thyroid nodules after RFA. Although there is no correlation between solitary or multiple nodules and whether they are benign or malignant, solitary nodules are an important prognostic factor. One study found a high recurrence rate after lobectomy for solitary nodules [19]. However, the internal mechanism of solitary nodules and their long-term prognosis has not been reported. We speculate that it may be related to the pathogenesis of nodules. In addition, the recurrence of nodules was associated with the type of surgery and the duration of follow-up [20]. Cesareo et al. [21] reported that nodule volume may be an important predictor of the efficacy of RFA in the clinical application of thyroid nodules. VPR of moderate nodules (>12ml) was significantly lower than that of the small nodule (>12ml), and the former had a higher risk of recurrence. Thyroid nodules usually show slow progression and slow growth rate, but the risk of postoperative recurrence is high, the difficulty of the second operation is increased, and the complications are also increased [22]. The larger the nodule diameter, the more complex the relationship between the nodule and the surrounding organs and blood vessels, which undoubtedly increases the risk of postoperative complications [23]. The adjacency of thyroid gland is complicated, and the distribution of blood vessels and nerves should be fully considered during the ablation. Therefore, for multiple nodules with large diameters, incomplete elimination may exist, which is one of the important reasons for the recurrence of thyroid nodules 24. In addition, nodular vasculature is significantly associated with thyroid nodule recurrence, and when nodules are visible or develop into vasculature, they have considerable regenerative potential [25, 26].

Zhao et al. [27] reported that RLN injury were associated with nodule diameter and the distance from nodule to thyroid sac. Poor recognition of recurrent laryngeal nerve and involuntary nerve excision during operation are the main causes of RLN injury [28]. Of course, these factors are related to nodule diameter, location and other factors. The occurrence of complications is related to the clinician's knowledge and manipulation, and the patient's condition. Whether the clinician is familiar with neck anatomy, ultrasound image interpretation, ablation power adjustment and ablation time control will affect the incidence of complications [29].

Whether a thyroid nodule needs surgical resection should be decided according to the size, nature and other factors of the nodule. RFA is suitable for the nodules to be benign and small, otherwise surgical resection may

be required. The former can eliminate thyroid nodules under the premise of small trauma and retaining thyroid function, while the latter has large trauma, wide resection and high complication risk. It has been reported that the recurrence rate of surgical resection is 29% to 42%, and preoperative volume of contralateral lobe and weight of resected thyroid gland can be important predictors of recurrence [30, 31]. And inadequate surgery is also one of the important factors for thyroid nodule recurrence. In addition, hypothyroidism is related to the method of excision, and hypothyroidism is obvious in patients with total thyroidectomy [31]. In contrast, RFA was better at preserving thyroid function in patients. Levothyroxine supplementation was not required in 28.3% of RFA patients one year after surgery [32]. Therefore, few studies use thyroid function as a predictor of recurrence. Nevertheless, RFA is an acceptable and effective treatment without increasing the risk of complications.

On the one hand, our study mainly has the following limitations: this research is a single-center retrospective analysis, with a small sample and possible sample bias. The conclusion is not representative. A multi-center and larger number of sample studies should be carried out to get more comprehensive evidence-based evidence. On the other hand, the short follow-up time of patients cannot predict the long-term disease recurrence of patients. Long-term follow-up should be carried out in the next study to further clarify the risk factors for disease recurrence after RFA treatment.

For patients with thyroid nodules, it is an important measure to improve the prognosis and reduce the economic burden of patients to improve the ability to predict postoperative recurrence of nodules and implement personalized prevention programs. This study showed that WNS, ND, DRN and recurrent laryngeal nerve injury were independent risk factors for the recurrence of thyroid nodules after radiofrequency ablation, and the combined model had high predictive value. Using this model, patients at high risk of recurrence can be screened as early as possible, so that they can be focused on and preventive measures can be taken.

Declaration of conflict of interest

All authors declare no conflict of interest.

Reference

1. Zamora EA, Khare S, Cassaro S. Thyroid Nodule. StatPearls. Treasure Island (FL): StatPearls Publishing Copyright © 2022, StatPearls Publishing LLC. 2022.

2. Venkatesh N, Ho JT. Investigating thyroid nodules. *Australian prescriber*. 2021;44(6):200-204.

3. Corrias A, Mussa A. Thyroid nodules in pediatrics: which ones can be left alone, which ones must be investigated, when and how. *Journal of clinical research in pediatric endocrinology*. 2013;5 Suppl 1(Suppl 1):57-69.

4. Kobaly K, Kim CS, Mandel SJ. Contemporary Management of Thyroid Nodules. *Annual review of medicine*. 2022;73:517-528 PubMed .

5. Zhang Z, Xia F, Li X. Ambulatory Endoscopic Thyroidectomy via a Chest-Breast Approach Has an Acceptable Safety Profile for Thyroid Nodule. *Frontiers in endocrinology*. 2021; 12:795627.

6. Sinclair CF, Bastien AJ, Ho AS. Radiofrequency ablation (RFA) compared to observation for benign thyroid nodules: emerging considerations for judicious use. *Current opinion in endocrinology, diabetes, and obesity*. 2022;29(5):449 PubMed -455.

7. Bellynda M, Kamil MR, Yarso KY. Radiofrequency ablation for benign thyroid nodule treatment: new solution in our center. *International journal of surgery case reports*. 2022; 97:107418.

8. Zhu Y, Zhang M, Jin Z, et al. Solid benign thyroid nodules (>10 ml): a retrospective study on the efficacy and safety of sonographically guided ethanol ablation combined with radiofrequency ablation. *International journal of hyperthermia: the official journal of European Society for Hyperthermic Oncology, North American Hyperthermia Group*. 2020;37(1):157-167.

9. Chen MH, Lin WC, Luo SD, et al. Residual, regrowth, and new growth of radiofrequency ablation for benign thyroid nodules of different volumes: two-year follow-up results. *International journal of hyperthermia: the official journal of European Society for Hyperthermic Oncology, North American Hyperthermia Group*. 2022;39(1):1172-1178.

10. Issa PP, Omar M, Issa CP, et al. Radiofrequency Ablation of Indeterminate Thyroid Nodules: The First North American Comparative Analysis. *International journal of molecular sciences*. 2022;23(19).

11. Bernardi S, Giudici F, Cesareo R, et al. Five-Year Results of Radiofrequency and Laser Ablation of Benign Thyroid Nodules: A Multicenter Study from the Italian Minimally Invasive Treatments of the Thyroid Group.

Thyroid: *official journal of the American Thyroid Association*. 2020;30(12):1759-1770.
12. Bernardi S, Cavallaro M, Colombin G, et al. Initial Ablation Ratio Predicts Volume Reduction and Retreatment After 5 Years from Radiofrequency Ablation of Benign Thyroid Nodules. *Frontiers in endocrinology*. 2020; 11:582550.
13. Bom WJ, Joosten FBM, van Borren M, et al. Radiofrequency ablation for symptomatic, non-functioning, thyroid nodules: a single-center learning curve. *Endocrine connections*. 2022;11(1): e210304.
14. Li L, Qiu X. Safety and Efficacy of Ultrasound-Guided Radiofrequency Ablation for Benign Non-functional Thyroid Nodules in Children: A Retrospective Study of 62 Patients with Over Four Years of Follow-Up. *Thyroid: official journal of the American Thyroid Association*. 2022;32(5):525-535.
15. Du JR, Li WH, Quan CH, et al. Long-term outcome of microwave ablation for benign thyroid nodules: Over 48-month follow-up study. *Frontiers in endocrinology*. 2022; 13:941137.
16. Radzina M, Cantisani V, Rauda M, et al. Update on ultrasound-guided radiofrequency ablation's role in thyroid nodule treatment. *International journal of surgery* (London, England). 2017;41 Suppl 1: S82-S93.
17. Familiar CC, Merino MS, Ganado DT, et al. Single-session treatment of benign thyroid nodules with radiofrequency ablation: Results at 6 months in 24 patients. *Endocrinologia, diabetes y nutricion*. 2020;67(3):164-171.
18. Lim JY, Kuo JH. Thyroid Nodule Radiofrequency Ablation: Complications and Clinical Follow Up. *Techniques in vascular and interventional radiology*. 2022;25(2):100824.
19. Marchesi M, Biffoni M, Faloci C, et al. High rate of recurrence after lobectomy for solitary thyroid nodule. *The European journal of surgery = Acta chirurgica*. 2002;168(7):397-400.
20. Boutzios G, Tsouroufl is G, Garoufalia Z, et al. Long-term sequelae of the less than total thyroidectomy procedures for benign thyroid nodular disease. *Endocrine*. 2019;63(2):247-251.
21. Cesareo R, Naciu AM, Iozzino M, et al. Nodule size as predictive factor of efficacy of radiofrequency ablation in treating autonomously functioning thyroid nodules. *International journal of hyperthermia: the official journal of European Society for Hyperthermic Oncology, North American Hyperthermia Group*. 2018;34(5):617-623.
22. Stan MN, Papaleontiou M, Schmitz JJ, et al. Nonsurgical Management of Thyroid Nodules: The Role of Ablative Therapies. *The Journal of clinical endocrinology and metabolism*. 2022;107(5):1417-1430.
23. Mauriello C, Marte G, Canfora A, et al. Bilateral benign multinodular goiter: What is the adequate surgical therapy? A review of literature. *International journal of surgery* (London, England). 2016;28 Suppl 1: S7-12.
24. Orloff LA, Noel JE, Stack BC, Jr., et al. Radiofrequency ablation and related ultrasound-guided ablation technologies for treatment of benign and malignant thyroid disease: An international multidisciplinary consensus statement of the American Head and Neck Society Endocrine Surgery Section with the Asia Pacific Society of Thyroid Surgery, Associazione Medici Endocrinologi, British Association of Endocrine and Thyroid Surgeons, European Thyroid Association, Italian Society of Endocrine Surgery Units, Korean Society of Thyroid Radiology, Latin American Thyroid Society, and Thyroid Nodules Therapies Association. *Head & neck*. 2022;44(3):633-660.
25. Offi C, Garberoglio S, Antonelli G, et al. The Ablation of Thyroid Nodule's Afferent Arteries Before Radiofrequency Ablation: Preliminary Data. *Frontiers in endocrinology*. 2020; 11:565000.
26. Baek JH. Factors related to the recurrence of benign thyroid nodules after thermal ablation. *International journal of hyperthermia: the official journal of European Society for Hyperthermic Oncology, North American Hyperthermia Group*. 2017;33(8):957-958.
27. Zhao ZL, Wei Y, Peng LL, et al. Recurrent Laryngeal Nerve Injury in Thermal Ablation of Thyroid Nodules-Risk Factors and Cause Analysis. *The Journal of clinical endocrinology and metabolism*. 2022;107(7): e2930-e2937.
28. Lenay-Pinon D, Biet-Hornstein A, Strunski V, et al. The circumstances in which recurrent laryngeal nerve palsy occurs after surgery for benign thyroid disease: a retrospective study of 1026 patients. *The Journal of laryngology and otology*. 2021;135(7):640-643.
29. Sim JS, Baek JH. Long-Term Outcomes of Thermal Ablation for Benign Thyroid Nodules: The Issue of Regrowth. *International journal of endocrinology*. 2021; 2021:9922509.
30. Lytrivi M, Kyrilli A, Burniat A, et al. Thyroid lobectomy is an effective option for unilateral benign nodular disease. *Clinical endocrinology*. 2016;85(4):602-608.
31. Yoldas T, Makay O, Icoz G, et al. Should subtotal thy-

roidectomy be abandoned in multinodular goiter patients from endemic regions requiring surgery? *International surgery*. 2015;100(1):9-14.
32. Yuan Q, Zheng L, Hou J, et al. Total thyroidectomy versus hemithyroidectomy with intraoperative radiofrequency ablation for unilateral thyroid cancer with contralateral nodules: A propensity score matching study. *Journal of otolaryngology - head & neck surgery = Le Journal d'oto-rhino-laryngologie et de chirurgie cervico-faciale*. 2022;51(1):26.

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Original Article

Comparison of the effectiveness of ultrasound-guided radiofrequency ablation and conventional open thyroidectomy in the treatment of benign thyroid nodules

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ABSTRACT

Objective: To compare the effectiveness of ultrasound (US)-guided radiofrequency ablation (RFA) and conventional open thyroidectomy (OT) in the treatment of benign thyroid nodules (BTN).

Methods: Medical records of 103 patients with BTN undergoing surgical treatment at The Affiliated Jiangning Hospital of Nanjing Medical University from March 2019 to March 2022 were retrospectively analyzed. Records show that 53 patients underwent US-guided RFA (observation group) and 50 patients underwent conventional OT (control group). Perioperative indicators (operation duration, intraoperative blood loss, postoperative hospital stay, incision length, and VAS score 12h and 24h after surgery), complications, thyroid function, and nodule recurrence in both groups were compared and analyzed.

Results: Perioperative indicators of patients in the observation group were better, and the visual analogue scale (VAS) scores at 12 and 24 hours after the surgery were lower than those of the control group ($p<0.05$). The incidence of complications in the observation group was significantly lower than that in the control group ($p<0.05$). There was no statistically significant difference in the preoperative levels of thyroid-stimulating hormone (TSH), serum free thyroxine (FT4) and serum free triiodothyronine (FT3) between the two groups ($p>0.05$). The postoperative TSH levels in the observation group increased compared to the preoperative levels and were higher than those in the control group, while FT4 and FT3 levels decreased after surgery and were lower than those in the control group ($p<0.05$).

Conclusions: Compared to conventional open thyroidectomy, US-guided RFA is associated with less trauma, faster recovery, fewer complications, and less impact on thyroid function in the treatment of patients with BTN.

KEYWORDS: Ultrasound-guided, Radiofrequency ablation, Conventional open thyroidectomy, benign thyroid nodules.

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INTRODUCTION

Benign thyroid nodules usually develop as a result of overgrowth of normal thyroid tissue, which accounts for over 90% of all identified thyroid nodules.[1] Although the prognosis of benign thyroid nodule (BTN) is good, they may lead to compression symptoms, such as voice changes, difficulty swallowing, and local protrusions may affect neck aesthetics.[1,2] Moreover, studies have shown that nearly 20% of BTN have a risk of subsequent malignancy.[3]

Surgery is a common treatment method for BTN. Open thyroidectomy (OT) is a routine procedure that involves removing nodules in an open state.[3,4] Although the therapeutic effect is reliable, the surgical trauma of OT is significant and may result in large scars.[4,5] Radiofrequency ablation (RFA) has emerged as a commonly used minimally invasive treatment method for BTN in recent years.[5,6] The thermal energy,

associated with radiofrequency ablation, causes localized protein coagulation, tumor cell necrosis, and gradual lesion absorption. RFA has the advantage of minimal trauma and fast recovery.[6,7]

In recent years, ultrasound (US)-guided RFA has been widely used in the treatment of BTN. A large number of clinical studies have shown that US-guided RFA is associated with over 80% postoperative nodule reduction rate after one year of follow-up, and over 90% postoperative nodule reduction rate after two years of follow-up.[8,9] However, there is currently a lack of clinical research comparing US-guided RFA with conventional OT in the treatment of BTN. This study aimed to compare and analyze the effects of US-guided RFA and conventional OT in the treatment of patients with BTN to provide reference suggestions for clinicians.

METHODS

Clinical records of 103 patients who underwent BTN resection at The Affiliated Jiangning Hospital of Nanjing Medical University from March 2019 to March 2022 were retrospectively analyzed. Among them, 53 patients underwent US-guided RFA were assigned to the observation group, and 50 patients underwent conventional OT were assigned to the control group.

Ethical Approval: The Medical Ethics Committee of our hospital approved this study (No. KY-2022-023, Date: Aug. 25th 2023).

Inclusion criteria:

- Met the diagnostic criteria for BTN, and BTN should be confirmed as benign by two separated fine-needle aspiration or core-needle biopsy.[10]
- Normal thyroid function indicators.
- Complete medical record and follow-up information.

Exclusion criteria:

- Malignant thyroid nodules.
- Bilateral thyroid nodules.
- Severe underlying diseases, combined with organ dysfunction.
- Presence of other throat diseases.
- Patients with a history of thyroid surgery.
- Patients with coagulopathy.
- The nodule is located behind the sternum.

Ultrasound-guided radiofrequency ablation: The instruments used were PHILIPS iU22 ultrasound system with probe L12-5 (Philips Medical Systems, Bothell, WA, USA) and multipole RFA instrument (LDRF-120S, Mianyang Lide Electronics Co., Ltd., Mianyang, China). Ablation was performed by an interventional consultant radiologist with six years' experience. After completing appropriate preoperative examinations, patient's posture was adjusted to supine position, the head was extended backward, fully exposing the neck. Neck infiltration anesthesia was initiated after routine disinfection and towel laying. Under ultrasound guidance, electrodes were inserted into the nodules and RFA was performed. During the treatment process, an appropriate amount of 0.9% sodium chloride was injected into the thyroid capsule to form a "liquid isolation zone". If the nodule was close to important tissues such as the laryngeal nerve, esophagus, etc., special attention was paid to protect the adjacent tissue. Complete ablation was not recommended. If the nodule contained cystic components or fluid, it was fully aspirated.

The ablation zone was determined according to the nodule condition shown by US examination. The power was maintained at 30~120W for 3~5s each time, and the ablation was generally completed within 40 minutes. After satisfactory ablation effect was achieved, absence of bleeding was confirmed, by the US, and desmopressin acetate was injected to stop the bleeding. The puncture site was treated, local compression was applied to stop the bleeding, and cold was applied to the skin for 30 minutes.

Conventional open thyroidectomy: OT was performed for all patients by the same surgical team. Patients were instructed to lie on their back. Pillows were placed on their shoulders to extend the neck and fully expose the surgical area. After routine disinfection and tissue laying treatment, an arc incision was made about 2cm above the sternum, and skin and subcutaneous tissues were dissected layer by layer until the thyroid was fully exposed along the midline of the neck. Based on the specific situation of the nodules, detected during the surgery, the mass was dissociated, completely removed, and bleeding stopped. A drainage tube was inserted, the incision sutured, and the surgery was completed.

Observation indicators:

- Perioperative indicators, including operation duration, intraoperative bleeding loss, postoperative hospital stay, incision length, and postoperative pain at 12 and 24 hours. Pain was evaluated using visual analogue scale (VAS). Scores are recorded by making a handwritten mark on a 10-cm line that represents a continuum between "no pain" and "worst pain."[11]
- The occurrence of surgical complications, including hematoma formation, hoarse voice, parathyroid injury, infection.
- TSH, TT4, and FT3 levels before and six months after the surgery. UniCel Dxl800 automatic electrochemiluminescence immunoassay (Beckman Coulter, USA) and matching reagents were used for detection.
- Recurrence of nodules 12 months after the surgery.

Statistical analysis: Data were put into Microsoft Excel and analyzed using SPSS version 26.0 (IBM Corp, Armonk, NY, USA). The normality of the data was evaluated using the Shapiro Wilk test. The data of normal distribution were expressed as mean ± standard deviation. Independent sample *t*-test was used for intergroup comparison, and paired *t*-test was used for intragroup comparison before and after the surgery. The data of non-normal distribution were expressed by

Table-I: Comparison of general data between the two groups.

Group	*Gender (Male/ Female)*	*Age (years)*	*Nodule volume(cm^3)*	*Nodule properties (n)*		
				Solid nodules	*Cystic nodules*	*Mixed nodules*
Control group (n=50)	26/24	46.2±7.8	3.51±0.69	26	17	7
Observation group (n=53)	35/18	47.4±8.9	3.65±0.80	18	26	9
χ^2/t	2.099	-0.726	-0.919		3.504	
p-value	0.147	0.470	0.360		0.173	

median and interquartile interval, and Mann Whitney *U* test was used for intergroup comparison. The counting data were represented by the number of use cases, and Chi-squared test was used. $p<0.05$ suggested that the difference is statistically significant.

RESULTS

A total of 103 patients were included in this study, including 61 males and 42 females. Age of the patients ranged from 32 to 68 years old, with an average of 46.8 ± 8.4 years. Patients (n=53) who underwent US-guided RFA comprised the observation group, and 50 patients who underwent traditional OT comprised the control group. There was no statistically significant difference in the baseline data between the two groups ($p>0.05$) (Table-I). The observation group had shorter operation duration and hospital stay compared to the control group, less intraoperative bleeding, shorter incision length, and lower VAS scores at 12 and 24 hours postoperatively ($p<0.05$) (Table-II).

The incidence of complications in the observation group was 1.9%, lower than the 12.0% in the control group ($p<0.05$) (Table-III). Before the surgery, there was no statistically significant difference in the levels of TSH, TT4, and FT3 between the two groups ($p>0.05$). Six months after the surgery, preoperative and postoperative levels of TSH, TT4, and FT3 were comparable in the observation group ($p>0.05$); However, the postoperative TSH level in the control group significantly increased compared to before the

Table-II: Comparison of operation conditions between the two groups.s

Group	*Operation duration (min)*	*Intraoperative blood loss (ml)*	*Postoperative hospital stay (d)*	*Incision length (mm)*	*VAS score 12h after surgery*	*VAS score 24h after surgery*
Control group (n=50)	49.6±10.6	17.4±2.5	4.80±1.14	28.7±2.9	6(5, 7)	4(3, 5)
Observation group(n=53)	25.8±4.3	4.37±1.33	1.51±0.41	2.65±0.40	3(3, 4)	2(2, 3)
t/Z	14.753	33.058	19.223	62.414	-7.498	-5.420
p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001

Table-III: Comparison of complication rate between the two groups.

Group	*Complication*				*Total*
	Hematoma formation	*Hoarseness*	*Parathyroid gland injury*	*Infection*	
Control group (n=50)	1 (2.0)	2 (4.0)	2 (4.0)	1 (2.0)	6 (12.0)
Observation group(n=53)	0 (0.0)	1 (1.9)	0 (0.0)	0 (0.0)	1 (1.9)
χ^2	-	-	-	-	4.154
p-value	-	-	-	-	0.042

Table-IV: Comparison of thyroid function between the two groups.

Group	TSH (mU/L)		FT4 (pmol/L)		FT3 (pmol/L)	
	Preoperative	6-months after surgery	Preoperative	6-months after surgery	Preoperative	6-months after surgery
Control group (n=50)	7.22±0.69	8.05±1.08*	18.55±1.98	16.31±1.83*	6.77±0.73	6.11±0.72*
Observation group(n=53)	7.16±0.73	7.24±1.02	18.83±2.23	18.67±2.10	6.62±0.67	6.53±0.67
t	0.411	3.922	-0.659	-6.064	1.067	-3.036
p-value	0.682	<0.001	0.512	<0.001	0.289	0.003

Note: Compared with preoperative data in this group, * p<0.05.

surgery and was higher than that in the observation group. Postoperative levels of TT4 and FT3 were lower than before the surgery and lower than those in the observation group (p<0.05) (Table-IV). Follow-up to 12 months after surgery showed no recurrence of nodules in both groups.

DISCUSSION

The results of this study show that compared with conventional open thyroidectomy, US-guided RFA has advantages of minimal trauma, faster recovery, fewer complications, and lower impact on thyroid function in patients with BTN, which is basically consistent with the findings by Yan et al.[12] Elhefny et al[13] also found that pain was more acceptable in RFA than surgery with few complications.

Numerous reports show that US-guided RFA approach in treating BTN is associated with less trauma, fewer complications, and can retain thyroid function.[9,14] The study by Jeong *et al*.[15] found that patients receiving RFA had shorter operation duration, shorter postoperative hospital stay, smaller incisions, less intraoperative bleeding, lower postoperative pain, and lower incidence of complications, which is consistent with our results. It is suggested that RFA has advantages such as minimal trauma, fast recovery, and fewer complications in the treatment of BTN. US-guided RFA is performed under real-time ultrasound guidance, and requires only puncture, without the need for a neck incision. Therefore, it can effectively shorten the surgical. In contrast, OT requires large incision for the removal of thyroid lobes, parathyroid glands, etc., and causes significant trauma with greater amount of intraoperative bleeding, prolonged postoperative recovery time, and increased risk of complications.[16,17] Fung *et al*.[18] conducted long-term follow-up on more than one thousand patients with thyroid nodules who received conventional open thyroidectomy, and found that about 28% of the patients had thyroid function damage after surgery, and required thyroxine, mainly due to the damage of parathyroid gland during surgery.[19]

In this study, patients who underwent US-guided RFA did not show significant changes in postoperative thyroid function, while patients who received conventional open thyroidectomy showed a significant decrease in thyroid function. Consistent with the study by Baldwin *et al*,[20] our results confirm the ability of US-guided RFA to further reduce thyroid function damage. US-guided RFA implements precise puncture through ultrasound guidance. During the ablation process, when the needle tip reaches a certain temperature, the process will automatically stop without causing damage to normal thyroid tissue. Additionally, a liquid isolation zone that is established by injecting saline into the thyroid capsule further prevents any damage that may potentially affect thyroid function.[21,22]

Our results showed that both groups of patients had no evidence of recurrence 12 months after the operation, which is consistent with the study by Guo *et al*.[23] Our results suggest that US-guided RFA can achieve similar surgical effects as conventional open thyroidectomy for BTN removal, and does not increase the rate of node recurrence. Taken together, our results confirm the efficiency of US-guided RFA, and suggest that this operation can be actively selected to further improve the clinical benefits of patients. It can provide a certain reference for clinical workers when undergoing thyroid surgery treatment.

***Limitations*:** This is a single-center retrospective study with a small sample size and short (only 12 months) follow-up after surgery, and only a few indicators were studied, which may affect the objectivity and extrapolation of our conclusions.

CONCLUSION

Compared with conventional OT, US-guided RFA has the advantages of less trauma, faster postoperative recovery, fewer complications, and less impact on thyroid function in the treatment of patients with BTN.

REFERENCES

1. Durante C, Costante G, Lucisano G, Bruno R, Meringolo D, Paciaroni A, et al. The natural history of benign thyroid nodules. JAMA. 2015;313(9):926-935. doi: 10.1001/jama.2015.0956
2. Akici M, Cilekar M, Yilmaz S, Arikan Y. Should intraoperative nerve monitoring be used routinely in primary thyroid surgeries? Pak J Med Sci. 2020;36(2):276-280. doi: 10.12669/pjms.36.2.1054
3. Ouyang S, Li W, Yu P, Li H, Cai H, Wu J. Effect of Chinese herbal medicine for patients with benign thyroid nodules in adults: A protocol for systematic review and meta-analysis. Medicine (Baltimore). 2021;100(8):e24591. doi: 10.1097/MD.0000000000024591
4. Sari R, Yabanoglu H, Hargura AS, Kus M, Arer IM. Outcomes of Total Parathyroidectomy with Autotransplantation versus Subtotal Parathyroidectomy Techniques for Secondary Hyperparathyroidism in Chronic Renal Failure. J Coll Physicians Surg Pak. 2020;30(1):18-22. doi: 10.29271/jcpsp.2020.01.18
5. Staniforth JUL, Erdirimanne S, Eslick GD. Thyroid carcinoma in Graves' disease: A meta-analysis. Int J Surg. 2016;27:118-125. doi: 10.1016/j.ijsu.2015.11.027
6. Xing Z, Qiu Y, Abuduwaili M, Xia B, Fei Y, Zhu J, et al. Surgical outcomes of different approaches in robotic assisted thyroidectomy for thyroid cancer: A systematic review and Bayesian network meta-analysis. Int J Surg. 2021;89:105941. doi: 10.1016/j.ijsu.2021.105941
7. Lee M, Baek JH, Suh CH, Chung SR, Choi YJ, Lee JH, et al. Clinical practice guidelines for radiofrequency ablation of benign thyroid nodules: a systematic review. Ultrasonography. 2021;40(2):256-264. doi: 10.14366/usg.20015
8. Suresh NV, De Ravin E, Barrette LX, Prasad A, Romeo D, Ng J, et al. Quality appraisal of clinical practice guidelines for the use of radiofrequency ablation in the treatment of thyroid nodules and cancer. Am J Otolaryngol. 2022;43(4):103508. doi: 10.1016/j.amjoto.2022.103508
9. Monpeyssen H, Alamri A, Ben Hamou A. Long-Term Results of Ultrasound-Guided Radiofrequency Ablation of Benign Thyroid Nodules: State of the Art and Future Perspectives-A Systematic Review. Front Endocrinol (Lausanne). 2021;12:622996. doi: 10.3389/fendo.2021.622996
10. Durante C, Hegedus L, Czarniecka A, Paschke R, Russ G, Schmitt F, et al. European Thyroid Association Clinical Practice Guidelines for thyroid nodule management. Eur Thyroid J. 2023;12(5):e230067. doi: 10.1530/ETJ-23-0067
11. Delgado DA, Lambert BS, Boutris N, McCulloch PC, Robbins AB, Moreno MR, et al. Validation of Digital Visual Analog Scale Pain Scoring with a Traditional Paper-based Visual Analog Scale in Adults. J Am Acad Orthop Surg Glob Res Rev. 2018;2(3):e088. doi: 10.5435/JAAOSGlobal-D-17-00088
12. Yan L, Li XY, Li Y, Luo Y. Ultrasound-Guided Radiofrequency Ablation versus Thyroidectomy for the Treatment of Benign Thyroid Nodules in Elderly Patients: A Propensity-Matched Cohort Study. Am J Neuroradiol. 2023;44(6):693-699. doi: 10.3174/ajnr.A7890
13. Elhefny AMM, El-Din AH, Abd-erRazik MA, Yassin A, Hussein A. Ultrasound-guided radiofrequency ablation versus total thyroidectomy for benign solitary thyroid nodules by fine-needle aspiration cytology. Egyptian J Sur. 2021;40(2):718-726. doi: 10.4103/ejs.ejs_75_21
14. Xue J, Teng D, Wang H. Efficacy and safety of ultrasound-guided radiofrequency ablation for papillary thyroid microcarcinoma: a systematic review and meta-analysis. Int J Hyperthermia. 2022;39(1):1300-1309. doi: 10.1080/02656736.2022.2129101
15. Jeong SY, Ha EJ, Baek JH, Kim TY, Lee YM, Lee JH, et al. Assessment of thyroid-specific quality of life in patients with benign symptomatic thyroid nodules treated with radiofrequency or ethanol ablation: a prospective multicenter study. Ultrasonography. 2022;41(1):204-211. doi: 10.14366/usg.21003
16. Yan J, Qiu T, Lu J, Wu Y, Yang Y. Microwave ablation induces a lower systemic stress response in patients than open surgery for treatment of benign thyroid nodules. Int J Hyperthermia. 2018;34(5):606-610. doi: 10.1080/02656736.2018.1427286
17. Wu X, Jiang Z, Liu J, Liu N, Hu Q, Xiong Y, et al. The efficacy and safety of microwave ablation versus conventional open surgery for the treatment of papillary thyroid microcarcinoma: a systematic review and meta-analysis. Gland Surg. 2022;11(6):1003-1014. doi: 10.21037/gs-22-243
18. Fung MMH, Lang BH. A prospective study evaluating the use of low-dose intravenous sedation and analgesia during radiofrequency ablation of symptomatic, benign thyroid nodules. Am J Surg. 2022;224(3):928-931. doi: 10.1016/j.amjsurg.2022.04.020
19. Yan L, Luo Y, Xiao J, Lin L. Non-enhanced ultrasound is not a satisfactory modality for measuring necrotic ablated volume after radiofrequency ablation of benign thyroid nodules: a comparison with contrast-enhanced ultrasound. Eur Radiol. 2021;31(5):3226-3236. doi: 10.1007/s00330-020-07398-0
20. Baldwin CK, Natter MB, Patel KN, Hodak SP. Minimally Invasive Techniques for the Management of Thyroid Nodules. Endocrinol Metab Clin North Am. 2022;51(2):323-349. doi: 10.1016/j.ecl.2022.01.001
21. Li L, Chen X, Li P, Liu Y, Ma X, Ye YQ. The Value of Ultrasound-Guided Fine-Needle Aspiration Cytology Combined with Puncture Feeling in the Diagnosis of Thyroid Nodules. Acta Cytol. 2021;65(5):368-376. doi: 10.1159/000517168
22. Yan L, Luo Y. Prediction of nodule regrowth after radiofrequency ablation of benign thyroid nodules. Int J Hyperthermia. 2021;38(1):11-12. doi: 10.1080/02656736.2020.1867243
23. Guo DM, Chen Z, Zhai YX, Su HH. Comparison of radiofrequency ablation and microwave ablation for benign thyroid nodules: A systematic review and meta-analysis. Clin Endocrinol (Oxf). 2021;95(1):187-196. doi: 10.1111/cen.14438

Authors' Contributions:

YF: Conceived and designed the study.
YX, HW and GZ: Collected the data and performed the analysis.
YF: Was involved in the writing of the manuscript and is responsible for the integrity of the study.
All authors have read and approved the final manuscript.